VISCERA

An exploration of touch, signal, decay, and our relationship to technology using mushrooms

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Project Overview

This project is a physical-digital artifact that explores how humans interact with technology **through touch rather than abstraction**.

The project exists as:

- A cinematic video artifact (“VISCERA”)
- A developing physical prop
- A conceptual system rooted in analog interaction

The final video functions as a **title sequence or artifact from a world in which the prop exists**, as the mushroom god itself is a work in progress.

What is a Mushroom God?

The project was inspired by an experiment in which pink oyster mushrooms were connected to a synthesizer, using their biological activity as a signal source. I wanted to explore this idea further by turning it into a physical artifact.

The mushrooms are grown inside a modified Bentley portable television. Electrical activity generated by water movement or physical touch is measured and used to trigger visual and mechanical responses, including a physical “magic 8 ball” that activates when a voltage threshold is reached. By becoming the most “capable” of all mushrooms, I have dubbed mine the Mushroom God. Pending completion.

Mushroom Selection

I needed to select for a couple of important qualities in whatever genus of mushroom I chose.

High Capacitance

Beginner friendly

Hardy

Accessible

I had initially started with Reishi, but due to slow growth rates, I switched to pink oysters, pictured to the right. Their growth was much faster.

Why is it relevant today?

With so much change in the world today, themes of alienation, the devaluation of the raw human experience, and the scarcity of physical touch were in mind when I started this project.

These include:

- Residual effects of social distancing
- The isolation of always being connected
- The lack of art designed for you to physically feel and experience like a“do not touch” policy at art exhibits of all sorts

The Central Question

What has an increasing lack of human contact changed about us?

The organic nature of the mushroom proposes an interface that:

- **Requires touch**
- **Responds to organic, unstable signals**
- **Isn't made of industrial materials**

Unlike the alienation of glass, plastic, rubber and metal, I want to create a real connection to a living being.

Why the Final Piece Is a Video

The video is not meant to represent the finished product. It is the first component of it.

Instead, it functions as:

- A title sequence
- A vessel for themes
- A cinematic introduction to the “world” it inhabits

This allowed me to:

- Establish tone and mythology
- Build a believable world
- Communicate intent while the physical prop is still evolving

The project is intended to continue beyond this semester, as I create the physical “prop” with real mushrooms and programmable computers.

My Style

My work is rooted in:

- **Black and white imagery**
- **Analog technology**
- **Film photography**
- **Engineering systems made visible, merging form and function**
- **The human experience of friction, uncertainty, and touch**

I am interested in the idea that most digital art is **experienced mechanically**, not physically — through screens, buttons, and interfaces that remove the body from the process.

Key Visual Influences

My visual language for Mushroom God draws heavily from:

- **Analog Film Photography**
- **Abstract Cinema**
- **Psychological Thrillers**
- **Obsolete technology**

Although my influences are not limited to these. Many of [Adult Swim]'s earlier works, as well as films produced by A24, and games such as Signalis, all had a great deal of influence on me

More Influences

Black & White as Constraint

The project is entirely black and white, with red subtitles for contrast.

This was a deliberate decision to:

- Keep in line with my other design work
- Emphasize form, texture, and contrast
- Avoid emotional cues associated with color

Black and white becomes a material, not a stylistic choice.

Typography & Colors

MD
NICHROME

#fdfcfe #0d0f10 #ff0e0e

Harsh two-tone contrast, red for subtitles.

What Is Dithering?

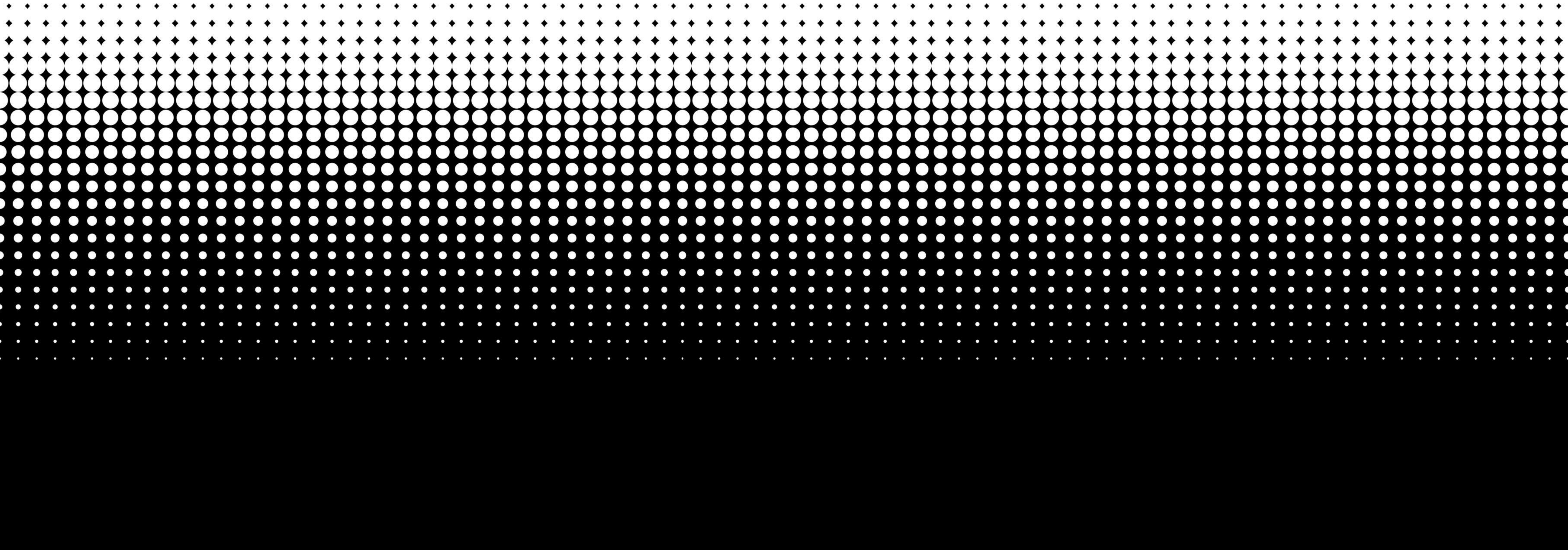
Dithering is a process where grayscale is simulated using patterns of pure black and white pixels.

There are no true gray values — only illusion.

Historically used in:

- Early computers
- Game Boy displays
- Low-resolution hardware

Example of Dotted Dithering



Technical Execution

I dithered my own photos in after effects to achieve the look of the video.

Threshold effects

Controlled dither scaling

Intentional loss of information

Rather than simulate modern clarity, I **destroyed resolution on purpose** to reveal structure and pattern.

> This process mirrors the project's philosophy: meaning emerges from reduction, not addition.

Retrofuturism & Physicality

Dithering represents:

- Technological limitation turned into expression
- A visual language born from constraint
- The most literal form of black and white that preserves shading

I see dithering as the **natural next step in black-and-white imagery:**

- No gradients
- No softness
- Plenty of texture

This aligns directly with the project's themes of raw signal and imperfect systems.

Sound as a Story

Audio sources include:

- Self-recorded audio (myself and a collaborator)
- Public domain recordings such as from the Apollo 17 mission

These were combined into a three act structure for Viscera

Act 1	Act 2	Act 3
Self Recorded	Apollo 17	The Conet Project

By strategically stitching together bits and pieces, I can create a storyline that makes sense while leaving room for the viewer’s imagination to fill in the gaps.The sound design reinforces the project's core themes: isolation, another layer of separation, and humanity's confrontation with systems beyond its control. I’m also HAM radio certified, which spared me much of the initial research.

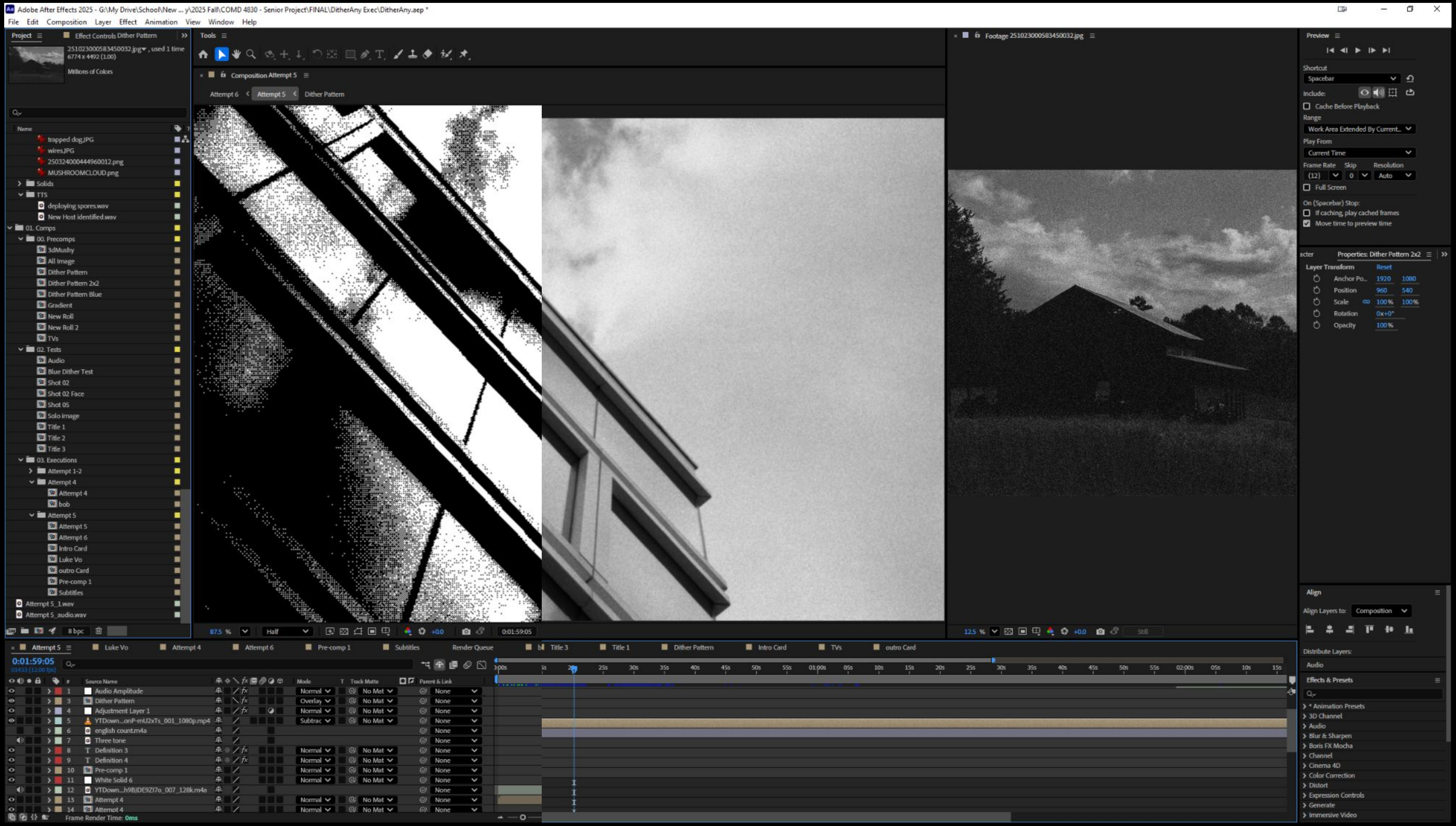
What Worked

A series of focused experiments were conducted, directly informing the final visual language and aesthetic of the project.

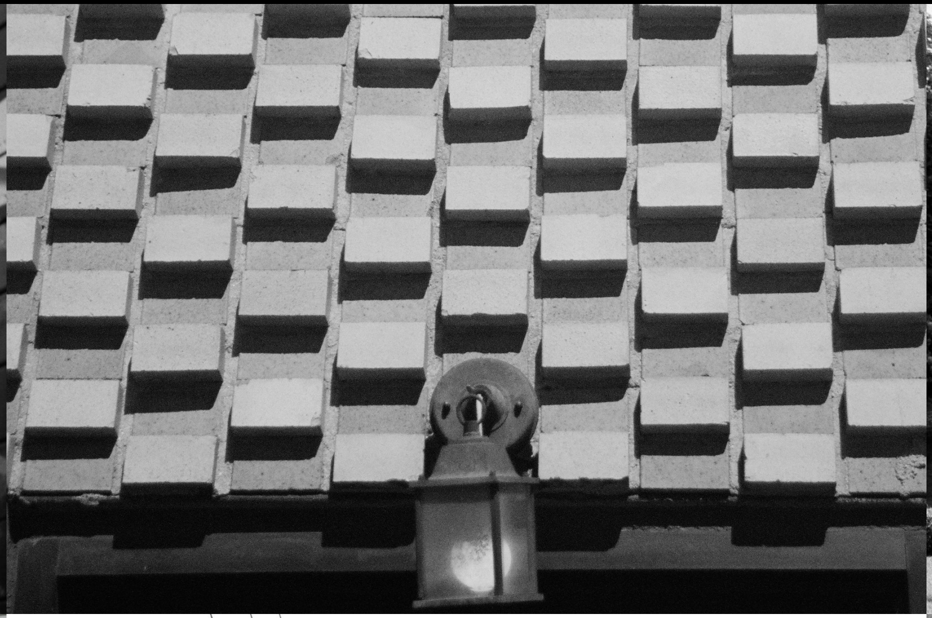
- **Dithering Pipelines**
- **Threshold Behaviors**
- **Noise & Degradation**
- **Photography stills**

These explorations were crucial in shaping the intentional destruction of resolution and the emphasis on raw, visible signal.

What Worked - Dithering



What Worked - Photography



What Didn't Work

Not all experiments were successful:

Recoloring experiments

Use of 3D Scans and Models

Using scanned rolls of film for footage

Whether they were visually appealing but betrayed the themes, or just plain didn't make sense, these are some of the things I tried that didn’t quite fit.

What Didn't Work

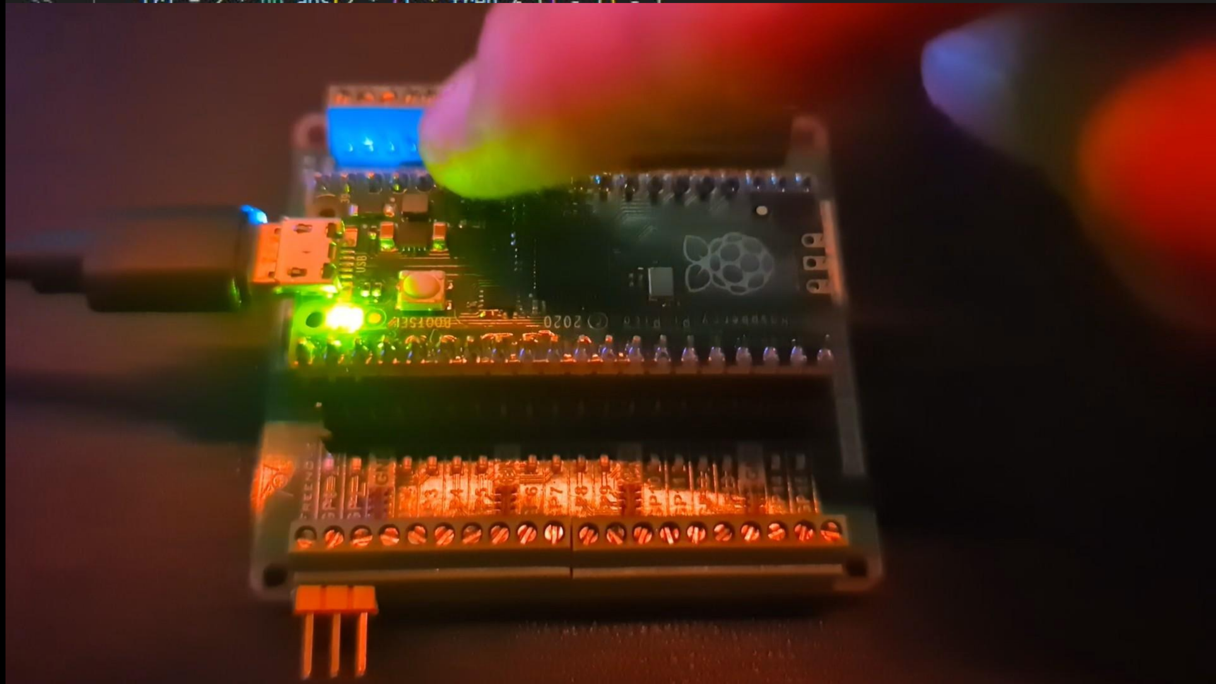
The scanned film, which is typically a favorite of mine, simply didn’t make sense with all the other visual language I had created with the world. The “technological era” everything else lived in was closer to early radio, CRT era technology. Showing the film seemed to me too analogue for the early-digital themes.

What Didn't Work

I was admittedly a little ambitious when first laying out the scope of my senior project, I had put quite a bit of time working on the prop itself in hopes that I may complete it before deadline. I prioritized the video regardless.

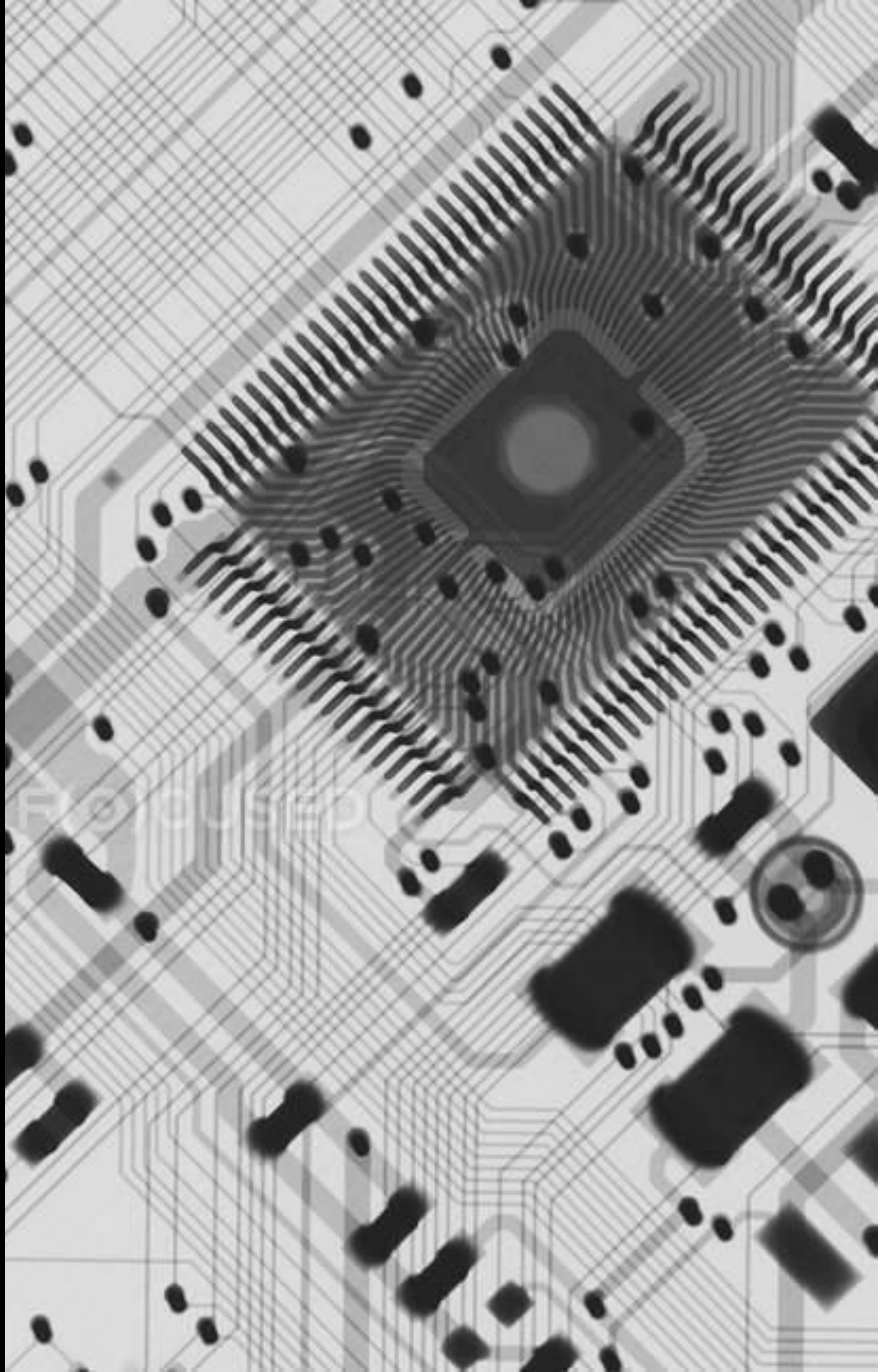
zero > oberon_synth.py > play_oracle_voice

```python
import pygame
import numpy as np
import time
import random

pygame.mixer.init(frequency=44100, size=-16, channels=2)
pygame.mixer.set_num_channels(8)

# Ritual scale
scale = [210, 234, 267, 311, 370, 392, 416]

def low_pass(signal, strength=0.05):
    filtered = np.copy(signal)
    for i in range(1, len(signal)):
        filtered[i] = filtered[i - 1] * strength + filtered[i] * (1 - strength)
    return filtered

def play_oracle_voice(freq, pan, delay=0.1, volume=0.9, bend_range=0.07, reverb=1.8):
    """
    OBERON's fungal whisper — warped, decaying tone with long tail and variable bend.
    """
    time.sleep(delay)

    sample_rate = 44100
    duration = 0.8
    t = np.linspace(0, duration, int(sample_rate * duration), endpoint=False)

    # Randomize bend intensity
    bend_dir = random.choice([-1, 1])
    bend_strength = random.uniform(0.0, bend_range)
    bend = 1 + bend_dir * bend_strength * (t / t[-1])**1.5

    # Organic waveform: sine + triangle + slow wobble
    base = np.sin(2 * np.pi * freq * bend * t)
```



So Who's This For?

- People who love mysteries, sci-fi and thrillers
- Those who enjoy unconventional, abstract stories
- Anyone who is philosophically aligned with what VISCERA tries to convey about modernity



Attribution

Due to constraints such as not being equipped for microscopic timelapses, or not being authorized to detonate a nuclear bomb, I’ve sourced public domain media.

Original Photography

The majority of photographic content in the project was my own, granting me direct artistic vision and control.

Sourced Mushroom Imagery

To overcome time and access limitations for capturing specific growth stages, select mushroom imagery was sourced.

Archival Imagery

Iconic mushroom cloud visuals, NSA patents, open-source 3D models and more were used in the production of VISCERA.

Every piece of sourced material was deliberately selected to support the project’s overarching themes where self-created media was unattainable.

Additional Resources

Name (hyperlinked)	Type	Notes
Mushroom Synthesizer	Inspiration/Research	Where it all started
Pepper’s Ghost effect	Inspiration/Research	For the Prop
Explore With Us	Inspiration	Video Production inspiration
Art of Dithering - Maxime Heckel	Research	Understanding how to dither
AARL	Research	Ham Radio association
La Jetee - Chris Marker	Inspiration	Black and white film, stills
Priom	Research	Numbers station archive
NASA Apollo 17 recordings	Audio	Act 2 - dialogue
The Conet Project	Audio	Act 3 - warning system
The Practical Zone system	Research	Book on maximizing contrast
Art of the Title	Inspiration	Mainly Saul Bass title sequences

Thank you for your time!

I look forward to hearing your feedback.